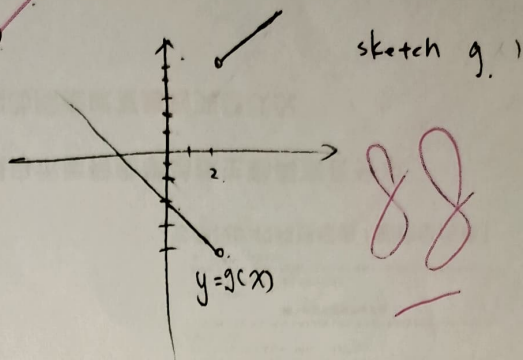


51. Let $g(x) = \frac{x^2+x-6}{|x-2|} + 2$

(a) $\lim_{x \rightarrow 2^+} g(x)$
 $= \lim_{x \rightarrow 2^+} \frac{(x+3)(x-2)}{x-2} + 2$
 $= \lim_{x \rightarrow 2^+} (x+3) + 2$
 $= 5$

(ii) $\lim_{x \rightarrow 2^-} g(x)$
 $= \lim_{x \rightarrow 2^-} \frac{(x+3)(x-2)}{-(x-2)} + 2$
 $= \lim_{x \rightarrow 2^-} -(x+3) + 2$
 $= -5$

(b) Does $\lim_{x \rightarrow 2} g(x)$ exist?
 (ans) **DNE**, $\lim_{x \rightarrow 2^+} g(x) \neq \lim_{x \rightarrow 2^-} g(x)$



2-5

51. $f(x) = \begin{cases} 0 & \text{if } x \text{ is rational} \\ 1 & \text{if } x \text{ is irrational} \end{cases}$

For what values of x is f continuous?

(ans) limit of x DNE at anywhere in $\mathbb{R}$, **None** of x is f continuous.

no specific number

2-6.

59.

f is a ratio of quadratic func.

has a vertical asymptote $x=4$,

just one x -intercept, $x=-1$

f has a removable discontinuity at $x=-1$

$\lim_{x \rightarrow -1} f(x) = 2$ $x \neq -1$

(a) $f(0)$

(ans)

$f(x) = \frac{a(x-1)(x+1)}{(x+1)(x-4)} = \frac{a(x-1)}{x-4}$

$\lim_{x \rightarrow -1} \frac{a(x-1)}{x-4} = 2 = \frac{-2a}{-5}$

$a = 5$

(b) $\lim_{x \rightarrow \infty} f(x)$

$\lim_{x \rightarrow \infty} \frac{(5x-5)^{\frac{1}{x}}}{(x-4)^{\frac{1}{x}}} = \frac{5 - \frac{5}{\infty}}{1 - \frac{4}{\infty}} = 5$

$f(0) = \frac{5(-1)}{-4} = \frac{5}{4}$

2-3.

42. prove that $\lim_{x \rightarrow 0^+} \sqrt{x} e^{(\sin \frac{x}{x})} = 0$

$$-1 \leq \sin \frac{\pi}{x} \leq 1$$

$$e^{-1} \leq e^{(\sin \frac{x}{x})} \leq e^1$$

$$\sqrt{x} e^{-1} \leq \sqrt{x} e^{(\sin \frac{x}{x})} \leq \sqrt{x} e$$

$$\lim_{x \rightarrow 0^+} \sqrt{x} e^{-1} \leq \lim_{x \rightarrow 0^+} \sqrt{x} e^{(\sin \frac{x}{x})} \leq \lim_{x \rightarrow 0^+} \sqrt{x} e$$

$$0 \leq \lim_{x \rightarrow 0^+} \sqrt{x} e^{(\sin \frac{x}{x})} \leq 0$$

$$\lim_{x \rightarrow 0^+} \sqrt{x} e^{(\sin \frac{x}{x})} = 0 \quad \text{by squeeze thm.}$$

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9/30 P.2.

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